

Functional Requirements Document (FRD)

Inventory Management Improvement System

Project	Inventory Management Improvement
Document	Functional Requirements Document (FRD)
Prepared By	Business Analyst
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Status	Draft

1. Purpose

This FRD translates the business requirements identified in the Inventory Management Improvement BRD into detailed functional requirements for an inventory-management solution. It describes what the proposed system should do, the information it should use, business rules, alerts, and expected user interactions.

2. Scope

In Scope

- Historical sales data analysis for demand forecasting.
- Current inventory monitoring.
- Product-level reorder point management.
- Reorder and production alerts.
- Slow-moving product identification.
- Inventory reporting and management visibility.
- User access for relevant business teams.

Out of Scope

- Automatic purchasing without business approval.
- Automatic production scheduling.
- Supplier contract management.
- Financial accounting and payment processing.

3. Users / Actors

Actor	Main Responsibility
Inventory Manager	Monitor stock, review alerts, manage reorder points, review slow-moving products.
Production Manager	Review replenishment alerts and decide production requirements.
Sales Manager	Review product availability and demand information.
Finance Manager	Review inventory cost and efficiency information.
Senior Management	Review KPIs, trends, and business-level inventory performance.

4. Functional Requirements

FR-01 — Historical Sales Data Import

Requirement: The system shall allow authorized users to load or connect historical sales data containing, at minimum, product, date, quantity sold, and sales value.

Business Rule / Note: The system must validate required fields before accepting the data.

FR-02 — Inventory Data Management

Requirement: The system shall maintain current inventory information for each product, including product ID, product name, category, current quantity, and inventory status.

Business Rule / Note: Inventory information must be identifiable by a unique product ID.

FR-03 — Demand Analysis

Requirement: The system shall analyze historical sales data to identify demand patterns by product and category.

Business Rule / Note: The system should support time-based analysis such as monthly or seasonal trends.

FR-04 — Demand Forecasting

Requirement: The system shall generate a demand forecast for each relevant product using available historical sales information.

Business Rule / Note: Forecast results should be viewable by product and time period.

FR-05 — Reorder Point Management

Requirement: The system shall allow authorized users to define and update a reorder point for each product.

Business Rule / Note: A reorder point must be greater than or equal to zero.

FR-06 — Inventory Monitoring

Requirement: The system shall regularly compare current inventory quantity against the defined reorder point.

Business Rule / Note: The comparison should be performed using the latest available inventory information.

FR-07 — Reorder Alert

Requirement: The system shall generate an alert when a product's inventory reaches or falls below its defined reorder point.

Business Rule / Note: The alert should identify the product, current stock, reorder point, and date/time of the alert.

FR-08 — Production Notification

Requirement: The system shall notify the Production team when a product reaches or falls below the defined replenishment threshold.

Business Rule / Note: The notification should be delivered within 1 minute of the threshold condition being detected.

FR-09 — Slow-Moving Product Identification

Requirement: The system shall identify products that meet the organization's defined slow-moving criteria.

Business Rule / Note: The slow-moving criteria must be configurable or documented by the business.

FR-10 — Inventory Dashboard / Reporting

Requirement: The system shall provide management with visibility into current stock, stockouts, reorder alerts, slow-moving products, and relevant inventory KPIs.

Business Rule / Note: Users should be able to filter the information by product, category, and time period where applicable.

FR-11 — Alert Status Tracking

Requirement: The system shall allow authorized users to view whether a replenishment alert is open, acknowledged, or resolved.

Business Rule / Note: The system should retain alert history for review.

FR-12 — Audit / Update History

Requirement: The system shall record important changes to reorder points and inventory-related settings.

Business Rule / Note: The record should include the user, change, and date/time.

5. Business Rules

- BR-01: A replenishment alert is triggered when current stock is less than or equal to the product's defined reorder point.
- BR-02: A product must have a valid product ID before it can be monitored.
- BR-03: The system must not automatically create a production order unless the business explicitly authorizes that capability.
- BR-04: Slow-moving status must be based on a business-defined period or sales threshold.
- BR-05: Only authorized users may change reorder points.
- BR-06: Inventory decisions should use the latest validated inventory data available.

6. Key Data Fields

Data Field	Purpose
Product ID	Unique product identification.
Product Name	Product description.
Category	Product classification.
Current Stock	Current available inventory quantity.
Reorder Point	Threshold at which replenishment should be considered.
Historical Quantity Sold	Input for demand analysis and forecasting.
Forecast Demand	Expected future demand.
Slow-Moving Status	Indicates whether the product meets the slow-moving criteria.
Alert Status	Tracks open, acknowledged, or resolved alerts.
Alert Date/Time	Records when an inventory condition was detected.

7. Main Process Flow

Load validated sales and inventory data → Analyze historical demand → Generate demand forecast → Set / review reorder points → Monitor current inventory → Compare current stock with reorder point → If threshold is reached, create alert → Notify responsible team → Production/Purchase decision → Replenish inventory → Update inventory → Monitor slow-moving products → Report KPIs to management.

8. Non-Functional Requirements

NFR-01 — Notification Response

The system should notify the Production team within 1 minute after detecting a replenishment threshold condition.

NFR-02 — Data Accuracy

Inventory information should be accurate and consistent enough to support reliable replenishment decisions.

NFR-03 — Availability

The inventory monitoring and reporting functions should be available during agreed business operating hours.

NFR-04 — Access Control

Only authorized users should be able to change reorder points or other controlled inventory settings.

9. Acceptance Criteria — Key Features

Feature	Acceptance Criteria
Demand Forecasting	Given valid historical sales data, when forecasting is run, then forecast results are generated and viewable by product/ti
Reorder Point	Given a product with a defined reorder point, when current stock reaches or falls below it, then the product is identified a
Reorder Alert	Given a threshold condition, when it is detected, then an alert is created containing product, stock, threshold, and timest
Production Notification	Given a valid replenishment alert, when it is created, then the Production team receives the notification within 1 minute.
Slow-Moving Products	Given the business-defined slow-moving criteria, when the criteria are met, then the product is identified in the report/da
Inventory Dashboard	Given authorized access, when a user opens the dashboard, then current inventory and key inventory KPIs are visible a

10. Assumptions

- Historical sales data is available and can be validated.
- Current inventory data can be captured with sufficient frequency for the monitoring process.
- The business will define appropriate reorder points and slow-moving criteria.
- Production and inventory teams have access to the required alerts.
- The final implementation technology will be selected separately.

11. Dependencies

- Availability and quality of sales and inventory data.
- Agreement on reorder-point calculation and ownership.
- Agreement on slow-moving inventory definition.
- Access and notification infrastructure.
- Management approval of business rules.

12. Traceability to BRD

BRD Requirement	Related FRs
BR-01 Reorder points	FR-05, FR-06, FR-07
BR-02 Demand forecasting	FR-03, FR-04
BR-03 Data-driven replenishment	FR-03, FR-05, FR-06, FR-07
BR-04 Slow-moving inventory	FR-09, FR-10
BR-05 Inventory visibility	FR-02, FR-06, FR-10, FR-11

13. Document Status

This FRD is a portfolio-level case-study document based on the assumptions and stakeholder findings established in the accompanying BRD. In a real project, the requirements would be validated and approved by the relevant stakeholders before development or implementation.